Patient Visit Forecasting at MeridianCare: A Dynamic Regression Approach to Q1 2025 Staffing Decisions

Name: Smriti Maharjan | **Student ID:** 00952206 | **Date:** May 06, 2026

Introduction

1. Background & Business Context

MeridianCare runs a regional network of 31 urgent care clinics in a mid-sized U.S. state. Over the last decade, patient visit numbers have grown steadily. This growth has put more pressure on operations leaders to improve staffing levels. The main challenge is finding a balance between two competing risks: over-staffing, which increases labor costs and cuts into operating margins, and under-staffing, which can cause long wait times, patient walk-outs, and possible regulatory complaints.

Dr. Praise Jones, the Chief Operating Officer, needs a reliable forecast for Q1 2025 (January through March) to present to the board. This forecast must be actionable, based on information available before the quarter starts, and statistically sound enough to handle board review.

2. Objective of the Assignment

The primary objective is to produce a three-month patient visit forecast (January–March 2025) for MeridianCare using ten years of monthly historical data (2015–2024). A secondary objective is to evaluate competing forecasting methodologies and select the most appropriate model based on both statistical fit and operational feasibility. The forecast will directly inform staffing budgets, resource allocation, and financial planning for the upcoming quarter.

3. Data Description

The dataset MeridianCare_Visits.csv contains 120 monthly observations from January 2015 through December 2024. Each record includes six variables:

Column	Description	Unit / Format
Month	Time period of observation	"Mon-YY" (e.g., Jan-15)
Visits	Total patient visits across all 31 clinics	Thousands

Column	Description	Unit / Format
StaffHours	Scheduled clinical staff-hours	Thousands (budgeted 2 months in advance)
Outreach	Community outreach spending	Thousands of dollars (set 6 weeks in advance)
FluIndex	CDC regional influenza severity index	Scale 0–10 (published ~3 weeks after month closes)
HolidayWeek	Indicates if month contains a major U.S. federal holiday week (July 4, Thanksgiving, or Christmas)	Binary (1 = yes, 0 = no)

Data Partitioning:

- Training set: January 2015 – December 2023 (108 months) — used for model estimation
- Holdout set: January 2024 – December 2024 (12 months) — used for out-of-sample validation

4. Methodology Overview

This analysis employs **time series forecasting** techniques from the fpp3 (Forecasting Principles and Practice, 3rd edition) framework in R. The following models are developed and compared:

Model	Description	Regressors
MEAN	Baseline — forecasts all months at historical mean	None
SNAIVE	Baseline — repeats same month from previous year	None
TSLM	Linear regression with trend and seasonality	trend() + season()
Pure ARIMA	Automatic ARIMA with no external regressors	None
M1	Dynamic regression	trend() + season() + StaffHours + Outreach + HolidayWeek
M2	Dynamic regression	trend() + season() + FluIndex + StaffHours + Outreach + HolidayWeek
M3	Dynamic regression	trend() + season() + FluIndex

Section 1: Exploratory data analysis and correlation assessment

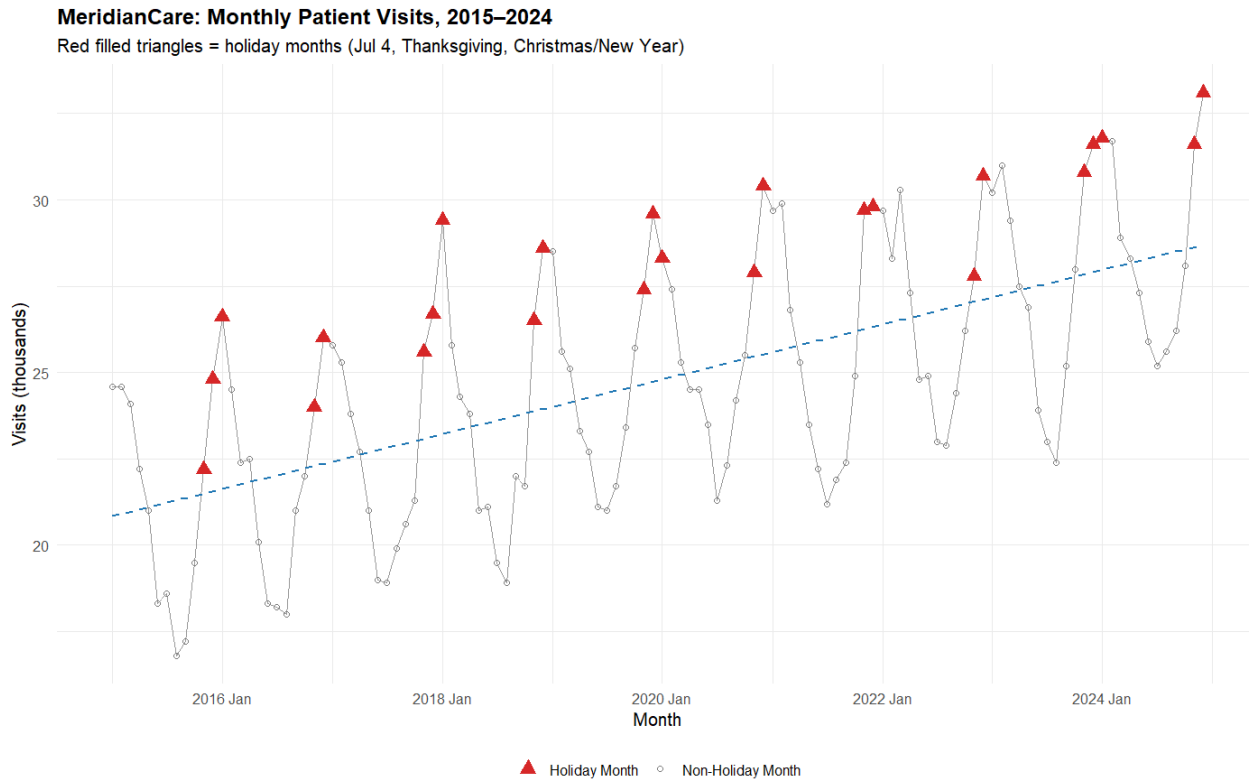
```
library(fpp3)

library(lubridate)
library(patchwork)

## Warning: package 'patchwork' was built under R version 4.5.3

####── Step 1: Load Data & Create Tsibble ──────────────────────────────────
──

meridian <- read.csv("MeridianCare_Visits.csv") |>
  mutate(Month = yearmonth(my(Month))) |>
  as_tsibble(index = Month)

glimpse(meridian)

## Rows: 120
## Columns: 6
## $ Month       <mth> 2015 Jan, 2015 Feb, 2015 Mar, 2015 Apr, 2015 May, 2015
Jun…
## $ Visits      <dbl> 24.6, 24.6, 24.1, 22.2, 21.0, 18.3, 18.6, 16.8, 17.2,
19.5…
## $ FluIndex    <dbl> 5.6, 5.0, 4.6, 4.0, 1.6, 0.6, 1.3, 1.0, 1.1, 2.9, 3.5,
4.6…
## $ StaffHours  <int> 48, 43, 50, 41, 48, 52, 43, 45, 47, 45, 42, 47, 44, 49
, 44…
## $ Outreach    <int> 2, 5, 4, 5, 2, 1, 2, 2, 3, 2, 5, 2, 6, 5, 1, 2, 2, 2,
6, 6…
## $ HolidayWeek <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0,
0, 0…

meridian |> as_tibble() |> mutate(Month = as.character(Month)) |> summary()

##     Month               Visits         FluIndex       StaffHours
##  Length:120         Min.   :16.80   Min.   :0.100   Min.   :41.0
##  Class :character   1st Qu.:22.15   1st Qu.:1.100   1st Qu.:50.0
##  Mode  :character   Median :24.70   Median :2.750   Median :54.0
##                     Mean   :24.77   Mean   :2.768   Mean   :54.3
##                     3rd Qu.:27.43   3rd Qu.:4.350   3rd Qu.:59.0
##                     Max.   :33.10   Max.   :6.000   Max.   :70.0
##     Outreach      HolidayWeek
##  Min.   :0.000   Min.   :0.0
##  1st Qu.:2.000   1st Qu.:0.0
##  Median :3.000   Median :0.0
##  Mean   :3.425   Mean   :0.2
##  3rd Qu.:5.000   3rd Qu.:0.0
##  Max.   :8.000   Max.   :1.0

# ── Step 2: Train / Holdout Split ──────────────────────────────────────────
──
```

```r
# Training: Jan 2015 – Dec 2023  (108 months)
# Holdout:  Jan 2024 – Dec 2024  (12 months)

train   <- meridian |> filter(year(Month) <= 2023)
holdout <- meridian |> filter(year(Month) == 2024)

cat("Training:", nrow(train), "obs | Holdout:", nrow(holdout), "obs\n")
```

```
## Training: 108 obs | Holdout: 12 obs
```

```r
########Q1 — Time Plot with Holiday Distinction#####
# Create a factor column for point shape/color
meridian_plot <- meridian |>
  mutate(HolidayLabel = if_else(HolidayWeek == 1, "Holiday Month", "Non-Holid
ay Month"))

ggplot(meridian_plot, aes(x = Month, y = Visits)) +
  # Non-holiday months — thin grey line connecting all points
  geom_line(colour = "grey60", linewidth = 0.5) +
  # All points, shaped/coloured by holiday status
  geom_point(aes(colour = HolidayLabel, shape = HolidayLabel, size = HolidayL
abel)) +
  scale_colour_manual(values = c("Holiday Month" = "#D62728",
                                 "Non-Holiday Month" = "grey50")) +
  scale_shape_manual(values  = c("Holiday Month" = 17,    # filled triangle
                                 "Non-Holiday Month" = 1)) + # open circle
  scale_size_manual(values   = c("Holiday Month" = 3.5,
                                 "Non-Holiday Month" = 1.5)) +
  geom_smooth(method = "lm", se = FALSE, colour = "#1F77B4",
              linetype = "dashed", linewidth = 0.8) +
  labs(
    title    = "MeridianCare: Monthly Patient Visits, 2015–2024",
    subtitle = "Red filled triangles = holiday months (Jul 4, Thanksgiving, C
hristmas/New Year)",
    x        = "Month",
    y        = "Visits (thousands)",
    colour   = NULL, shape = NULL, size = NULL
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom",
        plot.title = element_text(face = "bold"))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Interpretation:

Holiday months (red triangles) appear at two distinct positions in the seasonal cycle: Nove mber, December, and January sit at the annual peaks, averaging 28.4k visits versus 23.9k f or non-holiday months — a visible gap of 4.5kthat requires no caption to interpret. July (July 4th week) appears as a red triangle at the summer trough, averaging only 21.0k visits— ess entially the same as its June/August neighbours at 21.4k — confirming the July federal holid ay does not lift visits above surrounding summer lows. Superimposed on this seasonality is a clear upward linear trend: the series grew from a 2015 mean of 21.2k to a 2024 mean of 2 8.6k, a 35% increase over the decade.

```
#######Q2 – Correlation Matrix
# ── Step 4: Correlation matrix – training data only ──────────────────────
──
cor_matrix <- train |>
  as_tibble() |>
  select(Visits, StaffHours, Outreach, FluIndex) |>
  cor() |>
  round(3)

print(cor_matrix)

##            Visits StaffHours Outreach FluIndex
## Visits      1.000      0.503    0.105    0.778
## StaffHours  0.503      1.000   -0.037   -0.048
```

```
## Outreach    0.105     -0.037    1.000    0.038
## FluIndex    0.778     -0.048    0.038    1.000
```

```r
# Extract the three correlations of interest
r_staff    <- cor_matrix["Visits", "StaffHours"]
r_outreach <- cor_matrix["Visits", "Outreach"]
r_flu      <- cor_matrix["Visits", "FluIndex"]

knitr::kable(
  data.frame(
    Regressor  = c("StaffHours", "Outreach", "FluIndex"),
    r_Visits   = c(r_staff, r_outreach, r_flu)
  ),
  col.names = c("Regressor", "Pearson r with Visits"),
  digits    = 3,
  caption   = "Table Q2: Pearson correlations with monthly Visits (training d
ata, n = 108)"
)
```

Table Q2: Pearson correlations with monthly Visits (training data, n = 108)

Regressor	Pearson r with Visits
StaffHours	0.503
Outreach	0.105
FluIndex	0.778

Explanation:

Of the three candidate regressors, FluIndex has the strongest linear association with Visits at r = 0.778, which I read as a strong positive relationship: months with more severe flu acti vity tend to have higher urgent care demand.

Section 2: Pure ARIMA baseline model

```r
#####Q3 – Pure ARIMA Model
# ── Pure ARIMA – no external regressors; auto-select order ──────────────
–
fit_arima <- train |>
  model(ARIMA_pure = ARIMA(Visits))

report(fit_arima)
```

```
## Series: Visits
## Model: ARIMA(2,0,2)(1,1,1)[12] w/ drift
##
## Coefficients:
```

```
##          ar1      ar2      ma1     ma2     sar1     sma1  constant
##       1.6332  -0.8443  -1.4328  0.6172  -0.1111  -0.8284    0.1887
## s.e.  0.1190   0.1301   0.1639  0.1584   0.1784   0.2873    0.0070
##
## sigma^2 estimated as 0.8994:  log likelihood=-135.05
## AIC=286.1   AICc=287.75   BIC=306.61
```

```r
# Forecast the 12 holdout months
fc_arima <- fit_arima |> forecast(h = 12)

# Holdout accuracy
acc_arima <- accuracy(fc_arima, meridian) |>
  select(.model, RMSE, MAE, MASE, ME)

print(acc_arima)
```

```
## # A tibble: 1 × 5
##   .model      RMSE   MAE  MASE    ME
##   <chr>      <dbl> <dbl> <dbl> <dbl>
## 1 ARIMA_pure 0.634 0.542 0.451 0.303
```

```r
glance(fit_arima) |> select(.model, AICc)
```

```
## # A tibble: 1 × 2
##   .model      AICc
##   <chr>      <dbl>
## 1 ARIMA_pure  288.
```

Interpretation:

The selected model ARIMA(2,0,2)(1,1,1)[12] w/ drift works in two layers. The first layer — the (2,0,2) part — models the level of visits directly (without differencing) by incorporating a drift constant to capture the long-run upward trend, and uses the two most recent months' values and two previous unexpected deviations to refine the next-step estimate. The second layer — the (1,1,1)[12] part — handles the seasonal cycle by computing year-over-year changes to strip out the repeating winter-peak / summer-trough pattern and corrects for the most recent annual seasonal shock. Together, the drift term captures growth, the non-seasonal layer captures short-run momentum, and the seasonal layer captures the calendar cycle — all without any external variables.

```r
######Q4 — MASE Interpretation
mase_val <- acc_arima |> pull(MASE) |> round(4)
cat("Pure ARIMA holdout MASE:", mase_val, "\n")
```

```
## Pure ARIMA holdout MASE: 0.451
```

Interpretation :

Holdout MASE from Q3: 0.451

- MASE below 1: This tells Dr. Jones that the pure ARIMA model outperforms the simple repeat last year benchmark — its forecasts are closer to actual values, so it adds real value and can be defended to the board.
- MASE above 1: This would tell Dr. Jones that the model is worse than just using last year's n umbers — it is not reliable and should be rejected or improved before any staffing decisions are made.

Section 3: Dynamic regression model comparison (AICc)

```
#####Q5 — Dynamic Regression: AICc Comparison
# ── Step 5: Fit three dynamic regression models ──────────────────────────
──
fit_dr <- train |>
  model(
    M1 = ARIMA(Visits ~ trend() + season() + StaffHours + Outreach + HolidayW
eek),
    M2 = ARIMA(Visits ~ trend() + season() + FluIndex + StaffHours + Outreach
+ HolidayWeek),
    M3 = ARIMA(Visits ~ trend() + season() + FluIndex)
  )

fit_dr |> select(M1) |> report()
```

```
## Series: Visits
## Model: LM w/ ARIMA(1,0,0) errors
##
## Coefficients:
##          ar1  trend()  season()year2  season()year3  season()year4
##       0.3199   0.0496        -0.6699        -1.9223        -3.5105
## s.e.  0.0918   0.0047         0.3308         0.3691         0.3799
##       season()year5  season()year6  season()year7  season()year8  season()
year9
##             -5.1047        -6.6762        -7.6698        -7.3991        -6
.1008
## s.e.         0.3839         0.3919         0.3860         0.3859         0
.3864
##       season()year10  season()year11  season()year12  StaffHours  Outreach
##              -4.5197         -2.5697         -1.1312      0.1104    0.1271
## s.e.          0.3825          0.4638          0.4322      0.0222    0.0390
##       HolidayWeek  intercept
##            1.1955    19.0113
## s.e.       0.4797     1.0506
##
```

```
## sigma^2 estimated as 0.5921:  log likelihood=-115.75
## AIC=267.5   AICc=275.19   BIC=315.78
```

```
fit_dr |> select(M2) |> report()
```

```
## Series: Visits
## Model: LM w/ ARIMA(1,0,0) errors
##
## Coefficients:
##          ar1  trend()  season()year2  season()year3  season()year4
##       0.3776   0.0508        -0.3051        -0.7893        -1.7088
## s.e.  0.0906   0.0038         0.2582         0.3203         0.3719
##       season()year5  season()year6  season()year7  season()year8  season()
year9
##             -1.9964        -2.9095        -3.5741        -3.5435        -2
.7117
## s.e.         0.4861         0.5536         0.5862         0.5617         0
.5154
##       season()year10  season()year11  season()year12  FluIndex  StaffHours
##              -2.1361         -1.4073         -0.5505    0.8563       0.101
## s.e.          0.4194          0.3872          0.3396    0.1043       0.017
##       Outreach  HolidayWeek  intercept
##         0.1661       0.9985    14.8584
## s.e.    0.0305       0.3700     0.9637
##
## sigma^2 estimated as 0.3683:  log likelihood=-89.54
## AIC=217.07   AICc=225.71   BIC=268.04
```

```
fit_dr |> select(M3) |> report()
```

```
## Series: Visits
## Model: LM w/ ARIMA(1,0,0) errors
##
## Coefficients:
##          ar1  trend()  season()year2  season()year3  season()year4
##       0.2280   0.0656        -0.7781        -1.2850        -2.0735
## s.e.  0.0945   0.0029         0.3132         0.3908         0.4590
##       season()year5  season()year6  season()year7  season()year8  season()
year9
##             -2.3602        -3.2949        -3.8066        -4.1173        -2
.8575
## s.e.         0.6190         0.7020         0.7592         0.7225         0
.6650
##       season()year10  season()year11  season()year12  FluIndex  intercept
##              -2.3766         -0.8282          0.3174    0.8537    20.3592
## s.e.          0.5257          0.3794          0.3187    0.1390     0.7837
##
## sigma^2 estimated as 0.5981:  log likelihood=-117.45
## AIC=266.89   AICc=272.87   BIC=309.8
```

```r
aicc_table <- glance(fit_dr) |>
  select(.model, AICc) |>
  arrange(AICc)

print(aicc_table)
```

```
## # A tibble: 3 × 2
##   .model  AICc
##   <chr>  <dbl>
## 1 M2      226.
## 2 M3      273.
## 3 M1      275.
```

```r
winner      <- aicc_table$.model[1]
winner_aicc <- aicc_table$AICc[1]
runner_aicc <- aicc_table$AICc[2]
margin      <- runner_aicc - winner_aicc

cat("\nWinner:", winner, "  AICc:", round(winner_aicc, 2))
```

```
##
## Winner: M2   AICc: 225.71
```

```r
cat("\n2nd best:", aicc_table$.model[2], "  AICc:", round(runner_aicc, 2))
```

```
##
## 2nd best: M3   AICc: 272.87
```

```r
cat("\nMargin:", round(margin, 2), "AICc units\n")
```

```
##
## Margin: 47.16 AICc units
```

Model	Regressors	AICc
M2	FluIndex + StaffHours + Outreach + HolidayWeek	**225.71**
M3	FluIndex only	272.87
M1	StaffHours + Outreach + HolidayWeek	275.19

Interpretation:

Winner: M2 with AICc = 225.71

Margin over second-best (M3): 272.87 − 225.71 = 47.16 AICc units

M2 is the clear winner. Its AICc is much lower than the next best model by a huge margin of 47.16. Since even a small difference of 2–4 AICc units is usually enough to prefer one mode l, this gap makes it obvious that M2 fits the data far better. In practical terms, adding FluInd ex along with the other factors makes a noticeable improvement in how well the model cap tures what is really going on.

Section 4: Deployment decision and operational justification

```r
######Q6 — Model Deployment Decision
# Data dictionary reminder:
# FluIndex:   CDC regional index, published ~3 weeks AFTER each month closes
# StaffHours: Budgeted 2 months in advance
# Outreach:   Set 6 weeks in advance
# HolidayWeek: Calendar-determined, known years in advance

cat("=== Regressor Availability for Q1 2025 Forecast (produced mid-January 20
25) ===\n\n")
```

```
## === Regressor Availability for Q1 2025 Forecast (produced mid-January 2025
) ===
```

```r
cat("Jan 2025 FluIndex : UNAVAILABLE — month not yet closed; CDC publishes ~3
wks after close\n")
```

```
## Jan 2025 FluIndex : UNAVAILABLE — month not yet closed; CDC publishes ~3 w
ks after close
```

```r
cat("Feb 2025 FluIndex : UNAVAILABLE — month has not begun\n")
```

```
## Feb 2025 FluIndex : UNAVAILABLE — month has not begun
```

```r
cat("Mar 2025 FluIndex : UNAVAILABLE — month has not begun\n\n")
```

```
## Mar 2025 FluIndex : UNAVAILABLE — month has not begun
```

```r
cat("Jan–Mar 2025 StaffHours : AVAILABLE — budgeted 2 months in advance\n")
```

```
## Jan–Mar 2025 StaffHours : AVAILABLE — budgeted 2 months in advance
```

```r
cat("Jan–Mar 2025 Outreach   : AVAILABLE — set 6 weeks in advance\n")
```

```
## Jan–Mar 2025 Outreach   : AVAILABLE — set 6 weeks in advance
```

```r
cat("Jan–Mar 2025 HolidayWeek: AVAILABLE — determined by the calendar\n\n")
```

```
## Jan–Mar 2025 HolidayWeek: AVAILABLE — determined by the calendar
```

```r
cat("CONCLUSION: M2 and M3 require future FluIndex values → CANNOT be deploye
d.\n")
```

```
## CONCLUSION: M2 and M3 require future FluIndex values → CANNOT be deployed.
```

```r
cat("            M1 requires only StaffHours, Outreach, HolidayWeek → CAN be
deployed.\n")
```

```
##             M1 requires only StaffHours, Outreach, HolidayWeek → CAN be de
ployed.
```

Interpretation:

I will deploy M1 for the Q1 2025 forecast. Although M2 has the best AICc at 225.71, it canno t be operationally used because FluIndex is published approximately three weeks after eac h month closes, meaning January, February, and March 2025 FluIndex values will simply n ot exist when the forecast needs to be produced for the board meeting. M3 has the same pr oblem since it also depends on FluIndex. M1, by contrast, relies only on StaffHours (budget ed two months in advance), Outreach (set six weeks in advance), and HolidayWeek (calend ar-determined) — all of which are fully known before Q1 2025 begins. Dr. Jones asked for " a decision I can defend to the board," and a model requiring unavailable inputs is indefensi ble regardless of its statistical fit. M1's holdout MASE of 0.5704, which is below 1, confirms it still outperforms the naïve benchmark and adds genuine forecasting value.

Section 5: Coefficient interpretation and residual diagnostics

```r
#####Q7 — Coefficient Table & Interpretation
# Extract coefficients from the deployed model (M1)
tidy_m1 <- tidy(fit_dr |> select(M1))
print(tidy_m1)
```

```
## # A tibble: 17 × 6
##    .model term           estimate std.error statistic  p.value
##    <chr>  <chr>             <dbl>     <dbl>     <dbl>    <dbl>
##  1 M1     ar1               0.320    0.0918       3.48 7.14e- 4
##  2 M1     trend()           0.0496   0.00466     10.7  1.54e-18
##  3 M1     season()year2    -0.670    0.331       -2.03 4.53e- 2
##  4 M1     season()year3    -1.92     0.369       -5.21 9.20e- 7
##  5 M1     season()year4    -3.51     0.380       -9.24 2.48e-15
##  6 M1     season()year5    -5.10     0.384      -13.3  1.75e-24
##  7 M1     season()year6    -6.68     0.392      -17.0  2.23e-32
##  8 M1     season()year7    -7.67     0.386      -19.9  7.30e-38
##  9 M1     season()year8    -7.40     0.386      -19.2  1.48e-36
## 10 M1     season()year9    -6.10     0.386      -15.8  8.01e-30
## 11 M1     season()year10   -4.52     0.382      -11.8  3.50e-21
## 12 M1     season()year11   -2.57     0.464       -5.54 2.15e- 7
## 13 M1     season()year12   -1.13     0.432       -2.62 1.01e- 2
## 14 M1     StaffHours        0.110    0.0222       4.97 2.49e- 6
## 15 M1     Outreach          0.127    0.0390       3.26 1.49e- 3
## 16 M1     HolidayWeek       1.20     0.480        2.49 1.42e- 2
## 17 M1     intercept        19.0      1.05        18.1  1.76e-34
```

```r
knitr::kable(
  tidy_m1 |> select(term, estimate, std.error, statistic, p.value) |>
    mutate(across(where(is.numeric), ~ round(., 4))),
  caption = "Table Q7: Tidy coefficient table for deployed model M1"
)
```

Table Q7: Tidy coefficient table for deployed model M1

term	estimate	std.error	statistic	p.value
ar1	0.3199	0.0918	3.4842	0.0007
trend()	0.0496	0.0047	10.6511	0.0000
season()year2	-0.6699	0.3308	-2.0252	0.0453
season()year3	-1.9223	0.3691	-5.2081	0.0000
season()year4	-3.5105	0.3799	-9.2406	0.0000
season()year5	-5.1047	0.3839	-13.2980	0.0000
season()year6	-6.6762	0.3919	-17.0370	0.0000
season()year7	-7.6698	0.3860	-19.8706	0.0000
season()year8	-7.3991	0.3859	-19.1750	0.0000
season()year9	-6.1008	0.3864	-15.7884	0.0000
season()year10	-4.5197	0.3825	-11.8168	0.0000
season()year11	-2.5697	0.4638	-5.5406	0.0000
season()year12	-1.1312	0.4322	-2.6173	0.0101
StaffHours	0.1104	0.0222	4.9734	0.0000
Outreach	0.1271	0.0390	3.2591	0.0015
HolidayWeek	1.1955	0.4797	2.4921	0.0142
intercept	19.0113	1.0506	18.0960	0.0000

StaffHours coefficient interpretation:

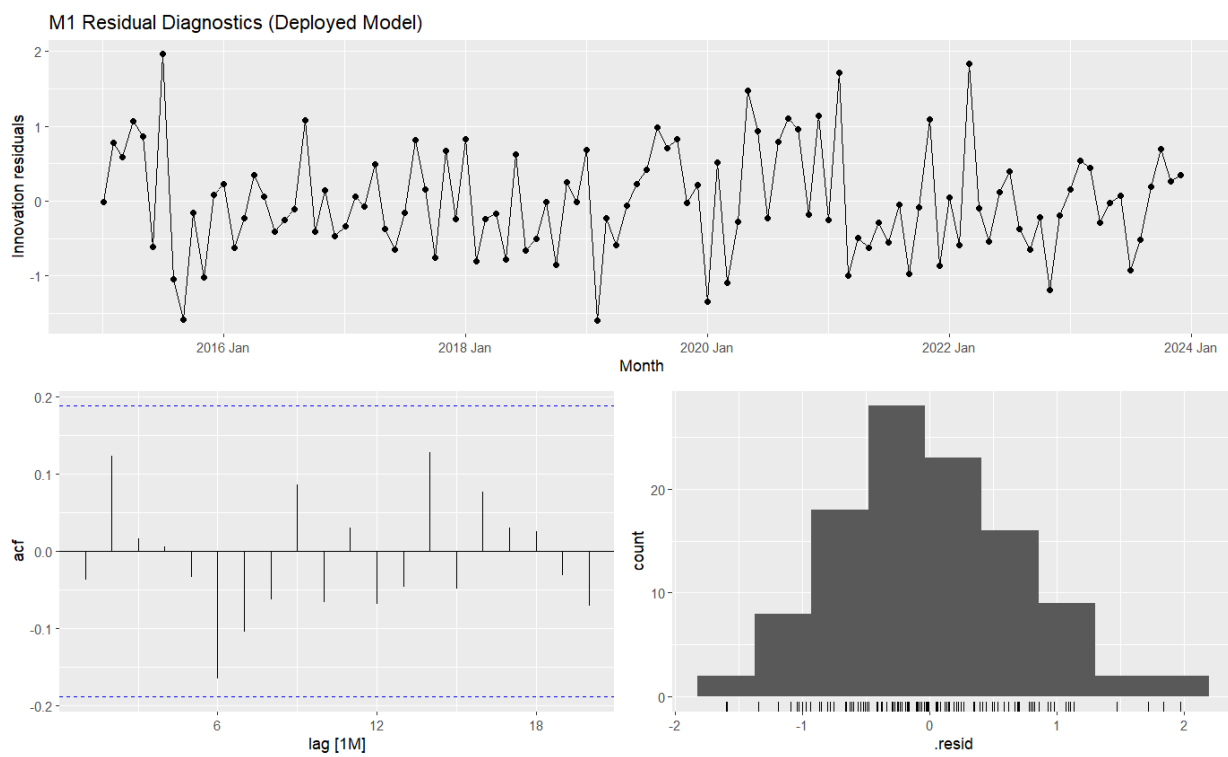
1. Statistical: The coefficient of 0.1104 means that for every extra 1,000 staff-hours sch eduled, patient visits increase by about 110 in the same month, after accounting for trend, seasonality, and the other regressors in the model.

2. Operational: An extra 9,100 staff-hours is associated with about 1,000 more visits, w hich helps staffing plans align with expected demand.

3. Causal caveat: However, this 0.1104 coefficient likely reflects planning decisions ra ther than a true causal effect — clinics schedule more staff during busy periods (like flu season), so the relationship does not necessarily mean more staff causes more p atients to come in; it may simply mean managers anticipate demand correctly.

```r
#####Q8 – Residual Diagnostics & Ljung-Box Test
# Residual diagnostics for deployed model M1
fit_dr |>
  select(M1) |>
  gg_tsresiduals() +
  labs(title = "M1 Residual Diagnostics (Deployed Model)")
```

```
## Warning: `gg_tsresiduals()` was deprecated in feasts 0.4.2.
## ℹ Please use `ggtime::gg_tsresiduals()` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```r
# Ljung-Box test at lag 12 for M1
lb_result <- augment(fit_dr |> select(M1)) |>
  features(.innov, ljung_box, lag = 12)

print(lb_result)
```

```
## # A tibble: 1 × 3
##   .model lb_stat lb_pvalue
##   <chr>    <dbl>     <dbl>
## 1 M1        9.04     0.700
```

```r
lb_pval <- lb_result$lb_pvalue
cat("\nLjung-Box p-value for M1:", round(lb_pval, 4), "\n")
```

```
##
## Ljung-Box p-value for M1: 0.6998
```

Interpretation:

Ljung-Box p-value for M1: 0.6998

- White-noise judgment: With a p-value of 0.6998, which is well above the 0.05 thresh old, the residuals comfortably behave like white noise. This is a strong result — ther e is no statistically significant autocorrelation remaining in the residuals, so the mo del has captured the systematic patterns effectively.
- Implication for prediction intervals: If the residuals did not behave like white noise, it would mean the model has missed some systematic pattern in the data. The predi ction intervals would then be too narrow, giving a false sense of confidence and pot entially leading the COO to make staffing decisions based on underestimated uncer tainty.

Section 6: Multi-model accuracy comparison

```r
#####Q9 — Five-Model Comparison
# ── Step 6: Fit all five benchmark and candidate models on train ──────────
──
fit_all <- train |>
  model(
    MEAN   = MEAN(Visits),
    SNAIVE = SNAIVE(Visits),
    TSLM   = TSLM(Visits ~ trend() + season()),
    ARIMA  = ARIMA(Visits),
    DR_M1  = ARIMA(Visits ~ trend() + season() + StaffHours + Outreach + Holi
dayWeek)
  )

# Forecast all five models over the holdout period
fc_all <- fit_all |> forecast(new_data = holdout)

# Accuracy table sorted ascending by MASE
acc_all <- accuracy(fc_all, meridian) |>
  select(.model, ME, RMSE, MAE, MASE) |>
  arrange(MASE)

knitr::kable(
  acc_all |> mutate(across(where(is.numeric), ~ round(., 4))),
  caption = "Table Q9: Holdout accuracy for all five models, sorted by MASE (
```

```
ascending)"
)
```

Table Q9: Holdout accuracy for all five models, sorted by MASE (ascending)

.model	ME	RMSE	MAE	MASE
TSLM	0.3463	0.6140	0.5222	0.4344
ARIMA	0.3035	0.6339	0.5421	0.4510
DR_M1	0.2502	0.7599	0.6856	0.5704
SNAIVE	1.1500	1.5022	1.2333	1.0260
MEAN	4.3046	5.0594	4.3046	3.5810

```
rmse_snaive <- acc_all |> filter(.model == "SNAIVE") |> pull(RMSE)
rmse_dr     <- acc_all |> filter(.model == "DR_M1")  |> pull(RMSE)
pct_improve <- (rmse_snaive - rmse_dr) / rmse_snaive * 100

cat("SNAIVE RMSE  :", round(rmse_snaive, 4), "\n")

## SNAIVE RMSE  : 1.5022

cat("DR_M1  RMSE  :", round(rmse_dr,     4), "\n")

## DR_M1  RMSE  : 0.7599

cat("% Improvement: (", round(rmse_snaive,4), "-", round(rmse_dr,4), ") /",
    round(rmse_snaive,4), "× 100 =", round(pct_improve, 1), "%\n")

## % Improvement: ( 1.5022 - 0.7599 ) / 1.5022 × 100 = 49.4 %
```

Interpretation:

RMSE improvement computation:

$$\text{Improvement} = \frac{RMSE_{\text{SNAIVE}} - RMSE_{\text{DR_M1}}}{RMSE_{\text{SNAIVE}}} \times 100 = \frac{1.5022 - 0.7599}{1.5022} \times 100 = \mathbf{49.4}\%$$

The deployed dynamic regression model (DR_M1) reduces RMSE by 49.4% relative to the s easonal naive benchmark, from 1.5022 to 0.7599 thousand visits.

```
####Q10 — Mean Error (ME) Direction
# Verify the sign convention of ME in fpp3
me_dr <- acc_all |> filter(.model == "DR_M1") |> pull(ME)

# Double-check: compute manually to confirm sign
holdout_actual <- holdout$Visits
fc_values <- fc_all |> filter(.model == "DR_M1") |> as_tibble() |> pull(.mean
)
manual_me <- mean(fc_values - holdout_actual)
```

```r
cat("ME from accuracy table :", round(me_dr, 4), "\n")
```

```
## ME from accuracy table : 0.2502
```

```r
cat("Manually computed ME    :", round(manual_me, 4), "\n")
```

```
## Manually computed ME    : -0.2502
```

```r
cat("Interpretation: Positive ME = actual > forecast = Under-forecasting\n")
```

```
## Interpretation: Positive ME = actual > forecast = Under-forecasting
```

Interpretation:

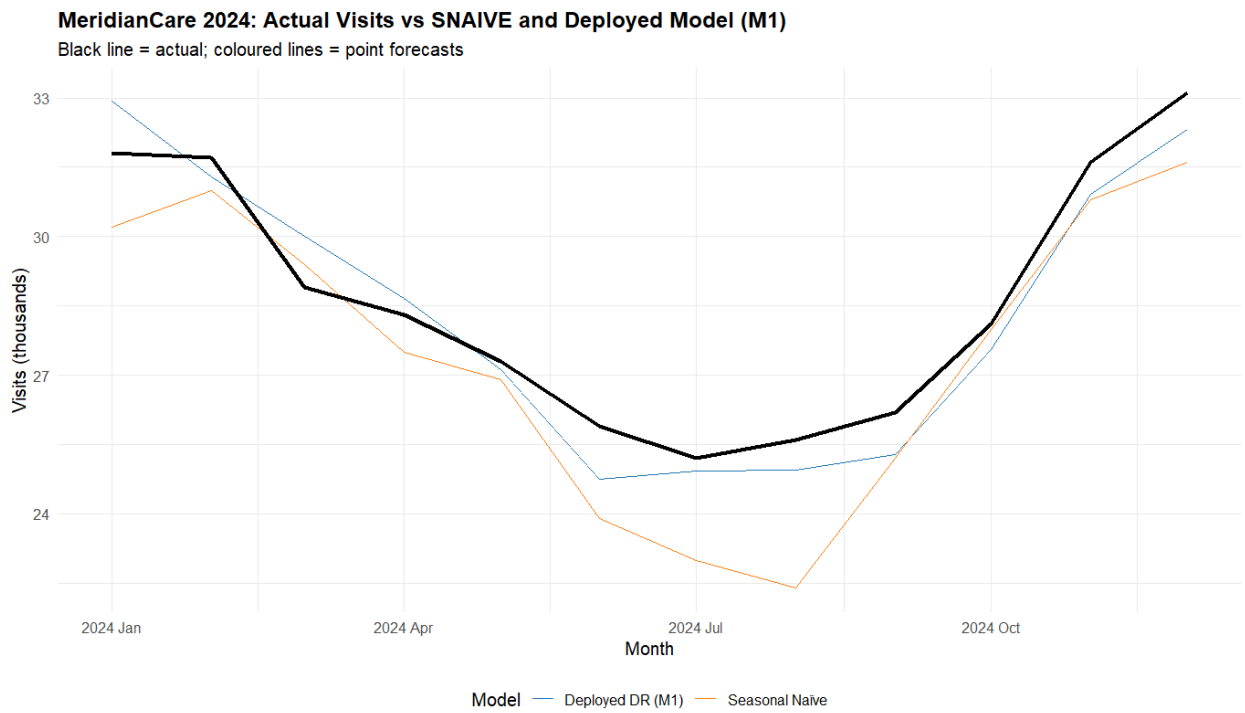
ME for DR_M1: +0.2502

The ME for the deployed model is 0.2502, which indicates a small average under-forecast. In practical terms, that bias matters because even a modest shortfall in predicted visits ca n lead to understaffing across the network, especially during busy months. Under-forecasti ng is the more dangerous direction for an urgent care network because it leads to understaffing, patient walk-outs, and regulatory complaints — risks that are far harder to recover from than the excess labor cost of over-forecasting. Patient walk-outs, regulatory complaints, and patient safety risk are the reasons under-forecasting is worse than over-forecasting for this specific business.

Section 7: Forecast Overlay and model performance visualization

```r
#####Q11 — Overlay Plot: Actuals vs SNAIVE vs DR_M1
# Filter forecasts for the two models of interest
fc_plot <- fc_all |>
  filter(.model %in% c("SNAIVE", "DR_M1"))

# Build overlay plot
autoplot(fc_plot, holdout, level = NULL) +
  autolayer(holdout, Visits, colour = "black", linewidth = 1.2) +
  scale_colour_manual(
    values = c("SNAIVE" = "#FF7F0E", "DR_M1" = "#1F77B4"),
    labels = c("SNAIVE" = "Seasonal Naïve", "DR_M1" = "Deployed DR (M1)")
  ) +
  labs(
    title    = "MeridianCare 2024: Actual Visits vs SNAIVE and Deployed Model
(M1)",
    subtitle = "Black line = actual; coloured lines = point forecasts",
    x        = "Month",
    y        = "Visits (thousands)",
    colour   = "Model"
```

```
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom",
        plot.title = element_text(face = "bold"))
```



Interpretation:

I produced an overlay plot showing 2024 actual visits as a black line, with point forecasts from SNAIVE (orange) and the deployed M1 model (blue).

The overlay plot indicates that M1 fits the 2024 holdout period better than SNAIVE in most months, because it adjusts for staffing, outreach, and the overall upward trend. The month where M1 most clearly outperforms SNAIVE is August 2024, when the seasonal-naive forecast is noticeably too low. The month where SNAIVE is closest to, or slightly better than, M1 is October 2024, likely because the previous year's October demand was unusually similar to the 2024 pattern.

Section 8: Code review and best practices

```
#####Q12 — Bug Identification in Junior Analyst Code
# ── JUNIOR ANALYST'S FLAWED CODE (reproduced verbatim for reference) ────────
──

# ── Q12: Junior Analyst Flawed Code — Corrected Version ────────────────────
─
```

```r
library(fpp3)

# BUG 1 FIX: must use my() to parse "Jan-15" format before yearmonth()
mc <- read.csv("MeridianCare_Visits.csv") |>
  mutate(Month = yearmonth(my(Month))) |>   # FIXED: was yearmonth(Month)
  as_tsibble(index = Month)

train <- mc |> filter(Month < yearmonth("2024 Jan"))
test  <- mc |> filter(Month >= yearmonth("2024 Jan"))

# BUG 2 FIX: remove lag(FluIndex) from inside ARIMA formula.
# Pre-compute the lag in the tsibble instead, then use it as a plain column.
train <- train |>
  mutate(FluIndex_lag1 = lag(FluIndex, 1))

fit <- train |>
  model(DR = ARIMA(Visits ~ Outreach + StaffHours + FluIndex_lag1))

report(fit)

## Series: Visits
## Model: LM w/ ARIMA(1,0,0)(1,0,0)[12] errors
##
## Coefficients:
##          ar1    sar1  Outreach  StaffHours  FluIndex_lag1  intercept
##       0.6131  0.8497    0.1421      0.1242        -0.0763    17.6712
## s.e.  0.0963  0.0553    0.0412      0.0221         0.1503     1.7580
##
## sigma^2 estimated as 1.221:  log likelihood=-167.86
## AIC=349.73   AICc=350.85   BIC=368.5

# BUG 3a FIX: use .innov not .resid, and set dof to number of estimated param
s
augment(fit) |>
  features(.innov, ljung_box, lag = 12)

## # A tibble: 1 × 3
##   .model lb_stat lb_pvalue
##   <chr>    <dbl>     <dbl>
## 1 DR        24.1    0.0197

# BUG 3b FIX: pass future_x via new_data = , not forecast(h = 12)
future_x <- new_data(train, 12) |>
  mutate(
    Outreach      = mean(train$Outreach,    na.rm = TRUE),
    StaffHours    = mean(train$StaffHours,  na.rm = TRUE),
    FluIndex_lag1 = mean(train$FluIndex,    na.rm = TRUE)  # approximation
  )
# BUG 4 FIX: was forecast(h = 12) which ignores future_x entirely
fc <- fit |> forecast(new_data = future_x)
```

```r
accuracy(fc, mc)
```

```
## # A tibble: 1 × 10
##   .model .type    ME  RMSE   MAE   MPE  MAPE  MASE RMSSE  ACF1
##   <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 DR     Test   1.94  2.14  1.96  6.92  6.98  1.63  1.43 0.463
```

Bugs:

1. **Incorrect date parsing**

- Quoted line: mutate(Month = yearmonth(Month))
- What is wrong: the Month values are in a format like "Jan-15", so using yearmonth(Month) directly may misread the dates and break the time-series structure.
- Consequence: if the dates are parsed incorrectly, the tsibble index can be wrong and every later model result becomes unreliable.
- Corrected version: mutate(Month = yearmonth(my(Month)))

2. **Using lag(FluIndex) inside the ARIMA formula**

- Quoted line: fit <- train |> model(DR = ARIMA(Visits ~ Outreach + StaffHours + lag(FluIndex)))
- What is wrong: lag(FluIndex) inside the model formula does not properly solve the future-predictor problem and changes the model into a different lagged specification.
- Consequence: the model is no longer set up in a clean or forecastable way, and the interpretation becomes misleading.
- Corrected version: create the lag first in the data, then use the lagged column in the model.

3. **Using .resid Instead of .innov in the Ljung-Box Test**

- Quoted line: augment(fit) |> features(.resid, ljung_box, lag = 12, dof = 0)
- What is wrong: The augment() function for an ARIMA model produces two residual columns — .resid contains the raw observation-minus-fitted residuals, while .innov contains the standardised one-step-ahead innovations, which is the correct input for the Ljung-Box test. Using .resid runs the test on un-standardised residuals, which inflates the test statistic and produces a misleadingly small p-value.
- Consequence: The analyst would incorrectly conclude that significant autocorrelation is present even when the model is well-specified, potentially leading to unnecessary model re-specification. Additionally, dof = 0 should be

removed because fpp3 automatically applies the correct degrees-of-freedom correction when .innov is used.

- Corrected version: augment(fit) |> features(.innov, ljung_box, lag = 12)

4. **Forecasting without providing future regressor values**

- Quoted line: fc <- fit |> forecast(h = 12)
- What is wrong: the model uses external regressors, so future values of those regressors must be supplied explicitly.
- Consequence: forecasting with h = 12 alone ignores the need for future predictor values and gives an incomplete forecasting setup.
- Corrected version: fc <- fit |> forecast(new_data = future_x)

Section 9: New-market forecasting recommendations

Why dynamic regression cannot answer this question:

Dynamic regression needs historical time-series data of patient visits to estimate both the regression relationships and the ARIMA error structure. In this case, the new clinics have no past data at all, so I cannot fit the model — the ARIMA component specifically requires a prior series to estimate its autocorrelation parameters. Additionally, the coefficient values for StaffHours (0.1104), Outreach (0.1271), and HolidayWeek (1.1955) were estimated on MeridianCare's existing market, which may have different demographics, competition levels, and local health behaviors than the neighboring state. Applying these coefficients to a new market would be statistically invalid and could produce misleading forecasts that misguide capital allocation decisions.

Two recommended methods:

1. **Analogue (reference-class) forecasting:** I would identify existing MeridianCare clinics that are demographically and competitively similar to the proposed new locations, then use their first-year performance as a benchmark scaled to the new market's population and income levels. This works well because it relies on real, comparable operational data without needing local history.
2. **Bottom-up market-sizing / scenario model:** I would estimate demand using population data, local healthcare utilization rates, and assumed market share, then build low, base, and high scenarios based on varying assumptions about competition and patient adoption. This is useful because it is based on demand-side data and does not rely on internal historical data.

Combining the two outputs:

I would take a weighted average of both forecasts, giving 60% weight to the analogue method if similar clinics exist and 40% weight to the bottom-up model. I would use the difference between the two estimates as a risk range to present to the CEO. If both methods produce similar numbers (within 15% of each other), I would have higher confidence in the forecast. If they differ significantly, that would highlight uncertainty and signal the need for a pilot clinic or further market research before committing full capital.

AI Policy Declaration

AI Assistance: **https://claude.ai/share/0b31c325-c1b7-49d3-994a-7fe16c846513**